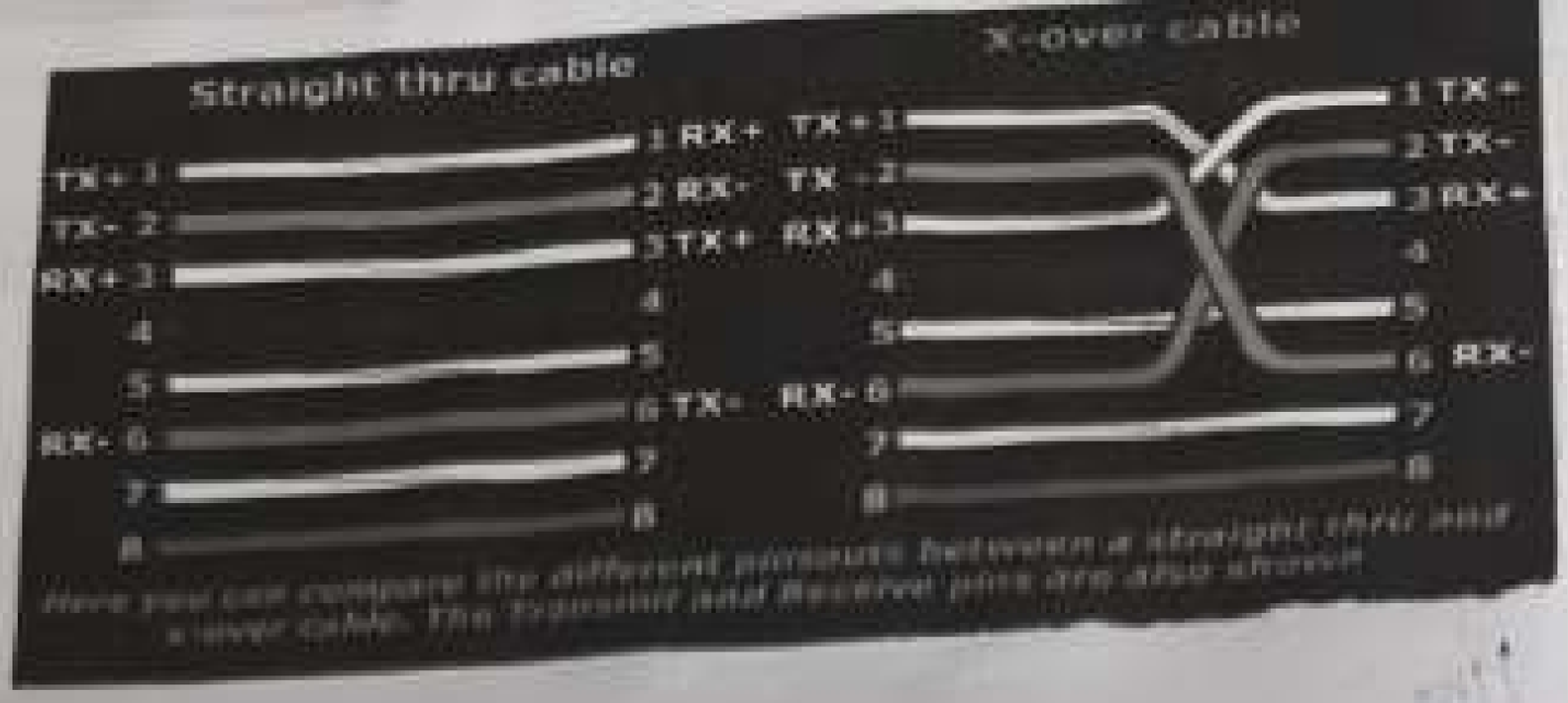


Here you can compare the different pinouts between a straight thru and x-over cable. The transmit and Receive pins are also shown.





Ex No. 02

Date. 23/01/25

Study of Different types of Network Cable

Aim:

To study and understand different types of network cable. To make your own Ethernet cross over cable / Straight Cable.

Different Types of Network Cable :-

1. unshielded twisted pair cable
2. Shield Twisted pair cable
3. Coaxial cable
4. Fibre optic cable.

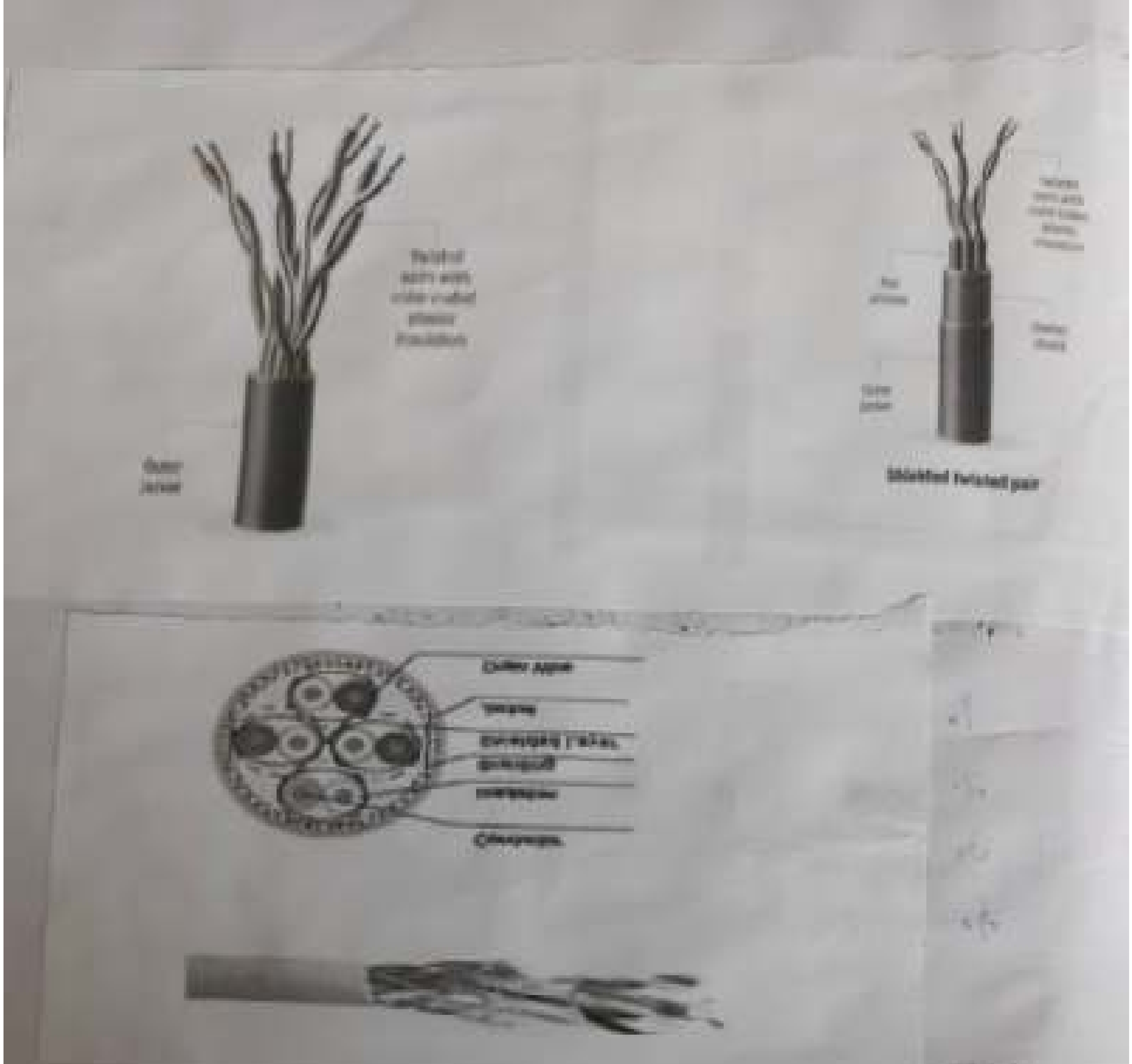
UTP :

* In Unshielded twisted pair cable has three categories

* Category 3 has 10Mbps Data Transmission Capa

* Category 6 has upto 100mbps Data Transmission Capacity

* Category 5e has 1Gbps has Capacity



Cable Type	Category	Maximum Data Transmission	Advantages / Disadvantages	Applications
UTP	Category 3	10Mbps upto 1000 mtrs	Advantages: * Cheaper in cost * Easy to install as they have small overall diameter Disadvantages * More prone to EMI	10 Base - T Ethernet
	Category 5			Fast Ethernet Gigabit Ethernet
	Category 5e	1Gbps		
STP	Category 6/6a	10Gbps	Advantages * Shielded * Faster than UTP * less prone to noise. Disadvantage * Expensive * Greater installation process	* Gigabit Ethernet * Widely used in data center
SSTP	Category 7	10 Gbps		* Gigabit Ethernet 10G Ethernet
Coaxial Cable	RG - 6 RG - 59 RG - 11	10 - 100 Mbps	Advantages * High bandwidth * Immune to interface * Versatile Disadvantage * Cost * Sizeable bulky	* Speed of Signal is 500m * High Speed internet Connec
Fibre Optics Cable	Single mode multimode	100Gbps	Advantages * High speed * High bandwidth * High Security Disadvantage * Expensive	* Maximum distance of fibre optics Cable is around ten


orange stripe
blue solid
white/blue stripe
Clip underneath,
pins facing you
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe

Make your own Ethernet Cross-over Cable / Straight Cable

Tools and Parts needed :-

* Ether Cabling Cat 5e is Certified for Gigabit Support but Cat5 Cabling works as well.
* A Crimping Tool
* Two RJ45 plugs
* Optional two plug shields

Step 1:-

To start Construction of the device, begin by Threading Shield on the table.

Step 2:-

To start Construction of next strip approximately 1.5 cm of the Cable Shielding from both ends

Step 3 :-

next you will need to untangle the wires there should be four "twisted pairs."

Step 4:-

Once the order is correct, bunch them together with a rubber band there are any then stick out from other.


white/green stripe
green solid
white/orange stripe
white/brown stripe

Step 5 :-

Next push the cable right in. The notch at end of the plug needs to be just over the cable shielding, and if it isn't, the means that you stripped off too much shielding. Simply snip the cables back a little more

Step 6 :-

After the wire are securely sitting inside the plug insert it into the Crimping tool and push down. It should be shaped correctly but put pushing too hard can crack the fragile plastic plug.

Step 7 :-

Lastly repeat for the other end using diagram B and diagram A.

To test it, plug it in and attempt to connect two devices directly.

Result :-

Thus, different type of network cables were studied successfully and Straight Through and Cross over Ethernet cables were prepared and tested successfully.